

Iris Flower Classification using Logistic Regression

Machine Learning Internship Project

1. Introduction

This project implements a machine learning classification model using the Iris dataset. The objective is to classify iris flowers into three species using Logistic Regression.

2. Dataset Description

The Iris dataset contains 150 samples and four input features: sepal length, sepal width, petal length, and petal width. The target contains three flower classes: Setosa, Versicolor, and Virginica.

3. Exploratory Data Analysis

Basic exploratory data analysis was performed to understand the dataset. The dataset was checked for missing values and statistical properties. Correlation between numerical features was also examined.

4. Feature Selection

All four numerical features were selected because they provide useful information for distinguishing the three Iris species.

5. Train-Test Split

The dataset was divided into training and testing sets using an 80/20 split. Stratified sampling was used to maintain the class distribution in both sets. There were 120 training samples and 30 testing samples.

6. Machine Learning Model

Logistic Regression was selected as the baseline classification model. The model was trained using the training dataset and then used to predict the classes of the test dataset.

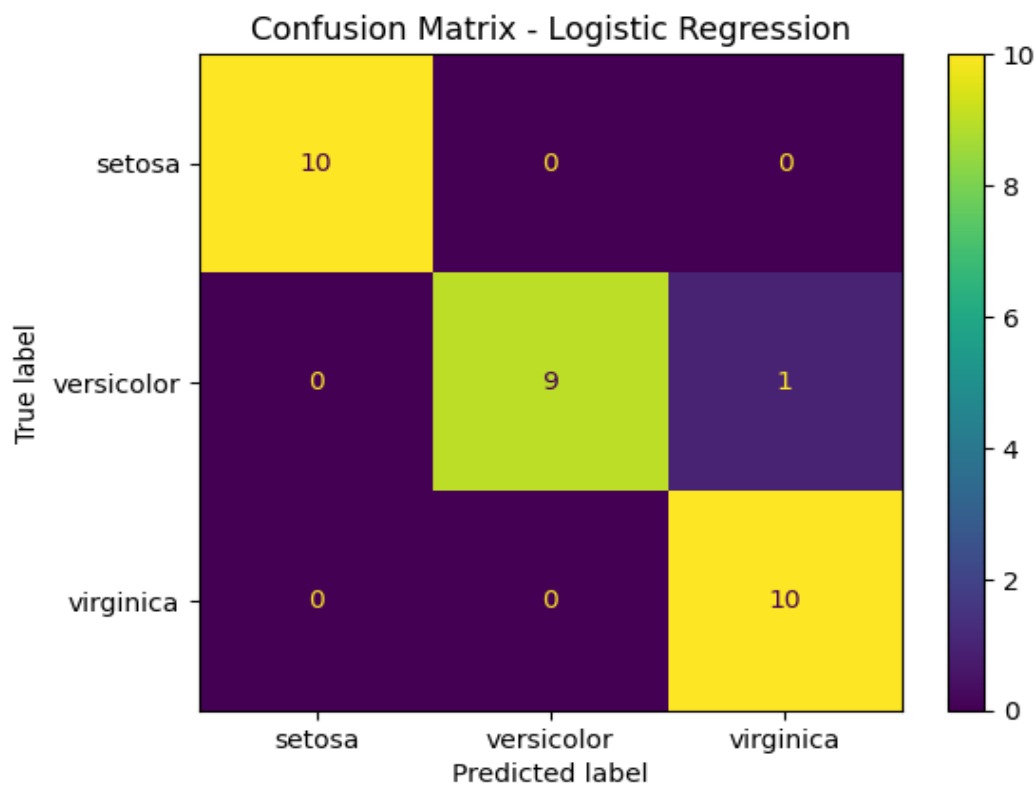
7. Model Performance

Model: Logistic Regression

Accuracy: 96.67%

Weighted F1-score: 96.66%

Confusion Matrix



8. Result Interpretation

The model correctly classified 29 out of 30 test samples. All Setosa and Virginica samples in the test set were correctly classified. One Versicolor sample was incorrectly classified as Virginica.

9. Limitations

The Iris dataset is small and relatively simple compared with real-world datasets. Therefore, the high performance obtained on this dataset may not represent performance on more complex datasets.

10. Potential Improvements

Future improvements could include comparing Logistic Regression with Decision Tree, Random Forest, and Support Vector Machine models. Cross-validation and hyperparameter tuning could also be applied to further evaluate and improve the model.

11. Conclusion

The Iris flower classification project successfully demonstrated a complete machine learning workflow including data loading, exploratory analysis, feature selection, train-test splitting, model training, and evaluation. Logistic Regression achieved 96.67% accuracy and a weighted F1-score of approximately 96.66% on the test dataset.